**Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology**
**(Deemed to be University Estd. u/s 3 of UGC Act, 1956)**

# WEEKLY LESSON PLAN

**Department:** Computer Science and Engineering

**Year / Semester:** 2025-26 / Winter

**Course Code / Course Name:** 10212CS292 / Applied Programming Skills

| **Unit III: Hierarchical and Priority-Oriented Data Structures**<br>**Course Outcome:** Apply tree and heap structures to handle hierarchical data and priority-based computations. | | | | |
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| **Week** | **Session** | **Topics to be Covered** | **Problems to Practice (Hands-On)** | **Platform** |
| Week 5 | Session 21 | Tree – Tree Traversal | Task 37: **Same Tree**<br>Task 38: **Symmetric Tree** | Leet Code |
| | Session 22 | | Task 39: **Binary Tree Inorder Traversal**<br>Task 40: **Binary Tree Preorder Traversal**<br>Task 41: **Binary Tree Postorder Traversal** | |
| | Session 23 | Print Birdseye View of a Tree | Task 42: **Vertical Order Traversal of a Binary Tree** | |
| | Session 24 | Path sum | Task 43: **Path Sum**<br>Task 44: **Path Sum II** | |
| | Session 25 | Root-to-leaf problems | Task 45: **Binary Tree Paths** | |

| Week | Session | Topics to be Covered | Problems to Practice (Hands-On) | Platform |
|---|---|---|---|---|
| Week 6 | Session 26 | Diameter/height problems | Task 46: https://takeuforward.org/data-structure/calculate-the-diameter-of-a-binary-tree<br>Task 47: https://takeuforward.org/data-structure/maximum-depth-of-a-binary-tree | takeuforward / Leet Code |
| | Session 27 | Lowest Common Ancestor | Task 48: **Lowest Common Ancestor of a Binary Tree**<br>Task 49: **Lowest Common Ancestor of a Binary Search Tree** | |
| | Session 28 | Kth smallest/largest in BST | Task 50: **Kth Smallest Element in a BST**<br>Task 51: **Kth Largest Element in an Array** | |
| | Session 29 | Heap Top-K pattern | Task 52: **Top K Frequent Elements** | |
| | Session 30 | K-way merge using heap | Task 53: **Find K Pairs with Smallest Sums** | |